

Design, Implementation, and Control of a Portable Clean-in-Place (CIP) System for an Industrial Machine, with Temperature Control, Monitoring, Traceability, and Cycle Analysis

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Abstract—Clean-In-Place (CIP) processes require controlled and repeatable sequences to reduce manual intervention and improve the traceability of process-related variables. This work presents the conceptual design of a portable CIP system intended to automatically execute the pre-rinse, washing, and final-rinse stages on a test system. The methodology includes preliminary characterization of the CIP process, formulation of verifiable requirements, evaluation of alternatives using Quality Function Deployment (QFD) and Pugh decision matrices, development of a conceptual mechatronic architecture through block diagrams, P&ID, and preliminary CAD design, as well as the definition of a thermal dynamic model, a preliminary temperature-control loop, and an industrial communication architecture for local and remote supervision. As a result of this first stage, the functional, control, instrumentation, communication, and validation foundations required to guide the subsequent construction and integration of the prototype were established. Full physical implementation, final PLC programming, final controller tuning, complete integration of the IoT layer, and experimental validation are planned for subsequent stages of the project.

Keywords—Clean-In-Place (CIP), portable system, process automation, temperature control, mechatronic design, industrial communications

I. INTRODUCTION

Clean-In-Place (CIP) cleaning enables the internal surfaces of tanks, pipelines, valves, and other process equipment to be cleaned without requiring complete disassembly, through the

circulation of water or cleaning solutions along their internal flow paths. Compared with manual cleaning procedures, this approach improves cycle repeatability, reduces direct operator intervention, and facilitates process traceability. In the context considered in this work, the cleaning cycle comprises three main stages: pre-rinse, washing with a cleaning solution, and final rinse.

The project originated from a technical visit to an industrial environment related to the food, beverage, and bottling sectors, where the need to investigate the automation of cleaning procedures applied to production equipment was identified. Based on this context, the development of an independent, modular, and portable solution was proposed, with the intention of making it applicable to different types of industrial equipment without reproducing the specific design of an existing machine.

Equipment used in these processes frequently lacks a portable unit capable of automatically and controllably executing the different stages of a CIP cycle directly on the production equipment, without relying on a fixed installation dedicated to a single production line. In the absence of such a unit, cleaning may be performed through manual or non-standardized procedures without temperature control, controlled dosing of the cleaning agent, or systematic recording of process variables. These conditions make it more difficult to ensure the repeatability, safety, and traceability required during

operation.

Based on this problem, the design question guiding this work is: how can a portable CIP cleaning system, applicable to different types of industrial equipment, be conceptually designed to automatically execute, control, and supervise a complete cleaning cycle within the time, budget, and scope constraints of an eighth-semester integrative project?

The relevance of this problem lies in the need to integrate different areas of mechatronics engineering into a single transportable solution. The system must consider cleaning-fluid preparation and handling, temperature control, instrumentation, sequence automation, and industrial communications while also enabling process supervision. From an industrial perspective, a modular and portable unit can be temporarily connected to different pieces of equipment without requiring a dedicated fixed CIP installation for each machine.

In response to this need, this work proposes the conceptual design of a portable CIP system for a test unit, integrating a mechatronic architecture, a preliminary temperature-control loop, and an industrial communication architecture intended for the automatic execution and supervision of the pre-rinse, washing, and final-rinse stages. The proposed design establishes the foundation for subsequent prototype construction, programming, integration, and experimental validation.

II. RELATED WORK

The background reviewed for this project focuses on two complementary technical areas: the development of Clean-In-Place processes and the integration of automation, industrial communication, and supervision technologies. This distinction is relevant because the proposed system is not limited to transporting cleaning fluids; it must also coordinate the CIP process with control, instrumentation, and communication functions.

In the field of CIP cleaning, Bowser [1] provides a general reference for understanding the Clean-In-Place principle, whereas Johansson [3] addresses the design and validation of a CIP process applied to a food-processing environment. In the project documentation, both references are used to support the programmed circulation of water, detergent solution, and rinse water, as well as the functional distinction among the pre-rinse, washing, and final-rinse stages. These studies provide the process-related foundation required to structure the cleaning cycle; however, their use within this project is primarily focused on CIP process characterization and validation.

The incorporation of industrial communications and supervisory functions introduces additional requirements related to control architecture and the exposure of industrial devices to external networks. Hajda, Jakuszczyk, and Ogonowski [9] analyze security challenges associated with PLC-based systems in Industry 4.0 environments. This issue is considered in the proposed system by maintaining the PLC as the central control element while establishing the connection to external services through an independent supervisory layer, thereby avoiding direct exposure of the controller to the Internet.

Regarding device interoperability, Cometa, Ambus, and Berjano [10] present an implementation of Modbus TCP communication between an ESP32 and a Siemens S7-1200 PLC. Complementarily, Martínez [11] documents the integration of an ESP32 with AWS IoT using MQTT. These studies provide references for establishing information-exchange mechanisms between industrial control devices and external supervisory systems. In this project, this type of integration is evaluated together with PROFINET for local PLC–HMI communication and alternatives based on Modbus TCP and MQTT for transmitting information to higher-level supervisory systems.

The combined analysis of these studies shows that the reviewed references provide specific solutions and technical foundations for different parts of the problem: [1], [3] address the CIP process, [9] addresses the security of connected industrial architectures, and [10], [11] address communication among PLCs, embedded devices, and supervisory platforms. However, within the set of studies considered in this project, these elements are predominantly addressed independently. None of these references, within the scope reviewed, presents a portable architecture that simultaneously integrates CIP cycle execution, cleaning-agent dosing, temperature control, PLC-based automation, and process-variable supervision.

Based on this technical gap, the present work proposes the conceptual integration of these subsystems within a modular and transportable CIP platform for a test unit. The contribution is therefore not the development of each technology individually, but rather their integration into a common mechatronic architecture intended to automatically execute and supervise the pre-rinse, washing, and final-rinse stages.

III. MATERIALS AND METHODS

A. General Development Methodology

The system was developed through a structured process of problem identification, analysis, and conceptual design. Initially, the team jointly defined the need to be addressed, the project scope, and the general conditions of the cleaning process to be automated. Based on this stage, a preliminary conceptual configuration of the system was established as a common foundation for the development of its different subsystems.

Subsequently, the activities were distributed according to the technical areas involved in the project, including CIP process characterization, requirements engineering, mechatronic and hydraulic design, temperature modeling and control, industrial communications, and formulation of the validation plan. The results obtained in each area were progressively integrated into a single conceptual proposal while maintaining consistency among the requirements, design decisions, and expected system functions.

Project documentation and technical developments were managed through a shared GitHub repository as a mechanism for version control and team integration. Modifications were incorporated through repository updates and recorded through commits, allowing a version history to be maintained and the

contributions made to the document and technical project files to be progressively integrated.

B. CIP Process Characterization and Requirements Engineering

The cleaning process was structured into three main stages: pre-rinse, washing with a detergent solution, and final rinse. For each stage, the fluid used, preliminary temperature and time conditions, involved actuators, transition criteria, and fluid destination were characterized. This characterization made it possible to define the general operating sequence, which subsequently served as a reference for the functional and control requirements.

Based on this information, a matrix of twenty verifiable requirements was formulated. The requirements covered process, functionality, portability, hydraulic system, electrical supply, electronics, instrumentation, control, industrial communications, IoT supervision, safety, and documentation. Each requirement was associated with an acceptance criterion, priority level, and verification method, thereby establishing a traceable relationship between system needs and subsequent design decisions.

The most relevant requirements included automatic execution of the complete cycle, temperature and level measurement, control of the fluid within the established thermal range, repeatable dosing of the cleaning agent, PLC–HMI communication, supervision of process variables, and safety conditions related to overtemperature and emergency stopping.

C. Conceptual Evaluation and Selection of Alternatives

Quality Function Deployment (QFD) was used to relate the identified needs to the technical characteristics of the system. This tool made it possible to organize the relationships between the requirements and the technical variables that influence system behavior, supporting the subsequent selection of the architecture, instrumentation, actuators, and control strategies.

Alternatives were compared using Pugh decision matrices. For each element, a reference alternative was established, and the remaining options were evaluated using a scale of +1 when their performance was superior, 0 when it was equivalent, and −1 when it was inferior to the reference.

Six general evaluation criteria were used: functional compliance, system compatibility, safety, cost and availability, maintainability, and ease of implementation. The weights assigned to these criteria were adjusted according to the function of the component or subsystem being evaluated.

Using this procedure, alternatives were compared for the hydraulic architecture, temperature sensor, level sensor, power stage, switching device, dosing system, agitation system, heating system, main pump, valves, communication protocols, and supervisory platform. Therefore, design decisions were not based solely on a single characteristic but on a weighted evaluation of multiple technical factors.

D. Conceptual Mechatronic Design

The mechatronic architecture was developed using a general block diagram, a functional architecture, a conceptual P&ID, and a preliminary CAD layout. The system was conceived as a portable unit capable of preparing, heating, dosing, and transporting the cleaning fluid to a test unit or test circuit.

The conceptual solution includes a main mixing tank, a tank for the cleaning agent, and a fluid-storage tank, together with an agitation system, heating element, main pump, dosing pump, and temperature and level instrumentation.

The conceptual P&ID shown in Fig. 1 identifies the main components using the equipment tags adopted in the project. TK-101 corresponds to the cleaning-agent tank, TK-102 to the mixing tank, and TK-103 to the storage tank. P-101 represents the dosing pump, while P-102 corresponds to the cleaning pump. The system also incorporates agitator AG-101, gearmotor M-101, heating element E-101, level transmitter LIT-101, and temperature transmitter TT-101.

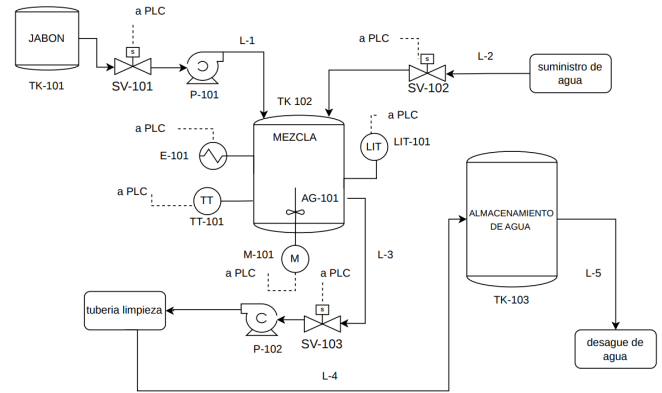


Fig. 1. Conceptual P&ID of the portable CIP system.

The CAD layout provided a preliminary verification of the location of the main components within a transportable structure. The design includes the main tank, agitation system, pump located in the lower section of the structure, and auxiliary containers intended for the cleaning agent and fluid storage. A general view of this arrangement is shown in Fig. 2.

Complementarily, Fig. 3 presents a top view of the CAD design, providing a clearer representation of the spatial arrangement of the main tank, auxiliary containers, agitation system, pumps, and instrumentation elements within the structure.

E. Conceptual Operating Sequence

Based on the defined architecture, a preliminary nine-stage sequence was established to represent system operation during a complete cleaning cycle.

The process begins by filling the main tank. Once the required level is reached, water heating is activated until the temperature defined by the control system is achieved. The fluid corresponding to the pre-rinse stage is then pumped through the test pipeline.

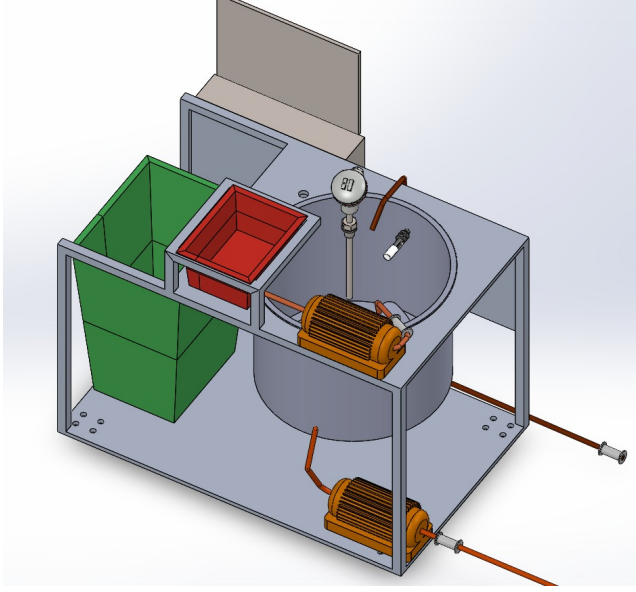


Fig. 2. General view of the conceptual CAD design of the portable CIP system.

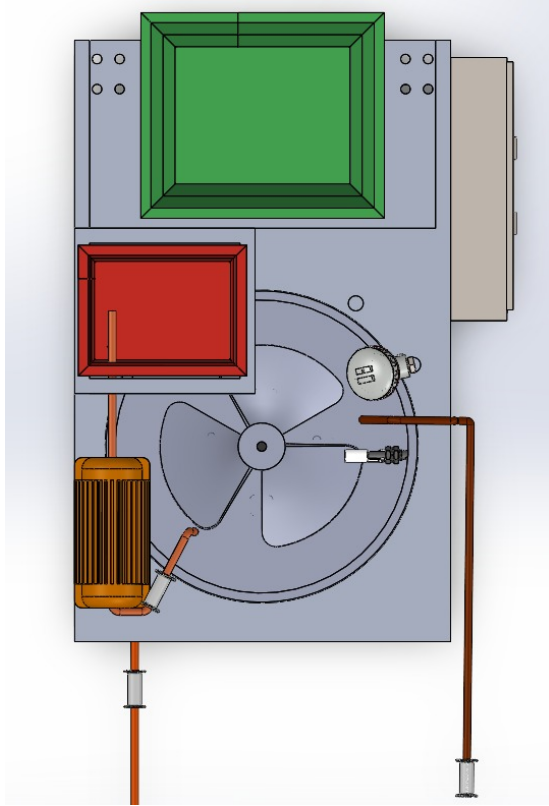


Fig. 3. Top view of the conceptual CAD design of the portable CIP system.

After the pre-rinse stage, the tank is refilled and the pump responsible for dosing the cleaning agent is activated. The mixture is homogenized by the agitator while the heating process corresponding to the washing stage is performed. The prepared solution is then pumped through the pipeline to be cleaned.

Finally, the tank is prepared again for the last stage of the cycle, corresponding to rinsing with clean water, whose purpose is to remove any remaining detergent and soil residues.

This sequence constitutes the functional basis for the subsequent automatic programming of the process in the PLC and preserves the general structure defined for the three main stages of the CIP cycle.

F. Thermal Modeling and Preliminary Control Strategy

The temperature of the fluid contained in the mixing tank was selected as the controlled variable of the system. Its measurement is proposed using a PT100 RTD sensor connected to the Siemens S7-1200 CPU 1214C PLC through the corresponding signal-conditioning interface. The manipulated variable is the average electrical power supplied to the 1500 W heating element.

To regulate this power, a time-proportional control strategy was proposed in which the PI-controller output determines the duty cycle applied to a zero-crossing solid-state relay (SSR). Controller execution is considered through the PID_Compact function block available in TIA Portal. The conceptual architecture of the control loop is presented in Fig. 4, where the temperature setpoint is compared with the measured temperature to generate the control error. The resulting signal acts on the heating element, while the process temperature is fed back to the controller. Variations in ambient temperature, thermal losses, and the addition of fluid are considered disturbances affecting the process.

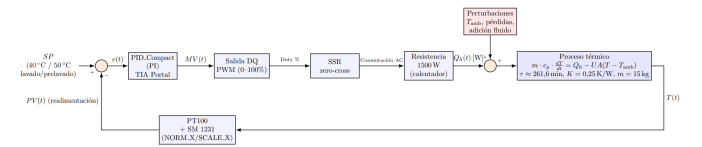


Fig. 4. Conceptual block diagram of the temperature-control loop for the CIP system.

To obtain a preliminary representation of the process dynamics, a first-order lumped-parameter thermal model was formulated. The energy balance considers the thermal power supplied by the heater, the energy stored in the fluid, and heat losses to the surrounding environment, according to

$$mc_p \frac{dT(t)}{dt} = Q_h(t) - UA [T(t) - T_{amb}], \quad (1)$$

where m is the fluid mass, c_p is its specific heat capacity, $T(t)$ is the fluid temperature, T_{amb} is the ambient temperature, $Q_h(t)$ is the supplied thermal power, and UA represents the overall thermal-loss coefficient.

By taking the temperature increase relative to ambient conditions as the output and the supplied thermal power as the input, the process behavior can be represented by the first-order transfer function

$$\frac{\Delta T(s)}{Q_h(s)} = \frac{K}{\tau s + 1}, \quad (2)$$

where the steady-state gain is defined as

$$K = \frac{1}{UA}, \quad (3)$$

and the thermal time constant is given by

$$\tau = \frac{mc_p}{UA}. \quad (4)$$

For the preliminary model, an active fluid mass of 15 kg and an approximate overall thermal-loss coefficient of 4 W/K were established. These parameters resulted in an approximate steady-state gain of 0.25 K/W and a time constant of 15697.5 s, equivalent to approximately 261.6 min. Consequently, the transfer function used to represent the thermal dynamics of the system was

$$G(s) = \frac{0.25}{15697.5s + 1}. \quad (5)$$

Based on this model, the open-loop thermal response of the system was analyzed using a maximum nominal power of 1500 W under different operating conditions. The simulation made it possible to observe the time evolution of the temperature and compare the process behavior for different power levels and temperature references. The resulting responses are presented in Fig. 5.

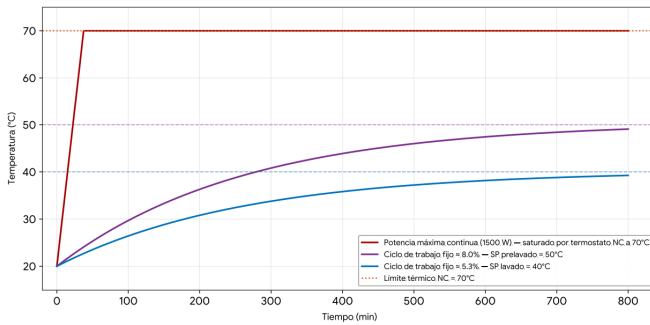


Fig. 5. Simulated open-loop thermal response of the system under different heating conditions.

The results showed the high thermal inertia of the process and indicated that operation based solely on a fixed power level cannot by itself guarantee accurate tracking of the desired temperature in the presence of thermal losses and disturbances. This behavior supports the use of a closed-loop control system. Therefore, a PI strategy with integral action and an anti-windup mechanism is proposed for subsequent stages to reduce steady-state error and limit the effects of actuator saturation during heating.

G. Industrial Communication Architecture

The communication architecture was developed with the PLC as the main node for acquisition and control of the system variables. Local communication between the PLC and the HMI was established using industrial Ethernet and PROFINET, taking advantage of the compatibility among the selected devices.

The conceptual topology combines a centralized structure around the PLC for control functions with an additional layer intended for information exchange with supervisory systems. As shown in Fig. 6, the process inputs and outputs are directly managed by the PLC, while the HMI provides local supervision. An ESP32-based gateway with a SIM800 module is incorporated into the communication network to provide an interface with external supervisory services. This organization keeps the main process functions within the local controller while separating remote supervision from critical automation functions.

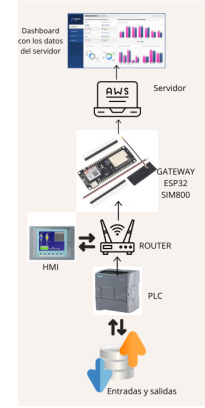


Fig. 6. Conceptual communication and supervisory architecture of the CIP system.

Different communication protocols were compared during the design stage, including S7, PROFINET, Modbus TCP, and OPC UA. MQTT was also considered as a mechanism for publishing variables to an IoT supervisory layer. In the proposed architecture, PROFINET is considered for local communication between the PLC and the HMI, while the gateway decouples communication with external services from the direct operation of the controller.

The upper supervisory layer considers the transmission of information to a remote server and its subsequent visualization through a dashboard, allowing relevant process variables to be accessed outside the local HMI. Therefore, the architecture is intended to support both local monitoring and the future availability of process information through remote supervisory platforms.

As functional evidence of the current project stage, communication between a Siemens S7-1200 PLC and an HMI was implemented through PROFINET, verifying the exchange and visualization of variables associated with the system status. Full integration of the gateway and remote supervisory layer is planned for subsequent stages of the project.

H. Validation Plan

The validation procedure was formulated through a requirement–test–evidence matrix. Each test was associated with one or more system requirements, and the experimental procedure, reference instrument, acceptance criterion, and type of evidence to be generated were defined.

The plan comprises twelve tests intended to evaluate automatic execution of the CIP sequence, valve actuation, temperature-control performance, overtemperature protection, dosing repeatability, absence of leakage, fluid recovery, emergency-stop operation, alarm generation, communications, level measurement, and mixture homogenization through the agitation system. The requirement–test–evidence matrix is presented sequentially in Tables I, II, and III.

TABLE I
REQUIREMENT–TEST–EVIDENCE MATRIX (1 OF 3).

ID	Req.	Procedure	Instrument	Criterion	Evidence
VAL-01	REQ01	Execute the complete automatic cycle (pre-rinse, washing, and final rinse) from the HMI without manual intervention in three consecutive tests.	Stopwatch, PLC log	3/3 complete executions in the correct sequence.	Cycle traces exported from the HMI.
VAL-02	REQ02	Verify the opening and closing of SV-101, SV-102, and SV-103 during each stage, confirming the absence of overlap between incompatible operating stages.	PLC digital-state log	Correct sequence without overlap in three consecutive tests.	Digital-variable traces.
VAL-03	REQ13	Set $SP = 50^{\circ}\text{C}$ for prewash and $SP = 40^{\circ}\text{C}$ for washing, and record the temperature response until steady state is reached.	TT-101 (PT100 + SM1231), reference thermometer	Steady-state error $\leq 2^{\circ}\text{C}$ with respect to the SP.	$T(t)$ versus SP graph and exported data.
VAL-04	REQ14	Deliberately induce a controlled overtemperature condition and verify shutdown of heater E-101 through the SSR.	PLC event log, stopwatch	SSR shutdown and warning activation when 60°C is exceeded.	Event log and HMI capture.

TABLE II
REQUIREMENT–TEST–EVIDENCE MATRIX (2 OF 3).

ID	Req.	Procedure	Instrument	Criterion	Evidence
VAL-05	REQ09	Use peristaltic pump P-101 to perform three repetitions with the same activation time and measure the volume of cleaning agent delivered in each test.	Graduated cylinder, stopwatch	Variation $\leq 10\%$ among the three measurements.	Photographic record and volume table.
VAL-06	REQ06	Execute a complete cycle and inspect tanks TK-101, TK-102, and TK-103 and their connections (SV-101, SV-102, SV-103, P-101, and P-102) for leakage.	Direct visual inspection	No visible leakage during the test.	Photographic record.
VAL-07	REQ06	Measure the volume of water supplied to the cycle through line L-2 and the volume recovered in storage tank TK-103 through line L-4 at the end of the cycle.	Graduated container or scale	Recovery percentage calculated and reported.	Volume and percentage calculation.
VAL-08	REQ18	Activate the emergency stop during cycle execution and measure the shutdown time of the pump, heater, and dosing system.	Stopwatch, PLC event log	Main actuators shut down within ≤ 1 s.	Signal capture or video.

TABLE III
REQUIREMENT–TEST–EVIDENCE MATRIX (3 OF 3).

ID	Req.	Procedure	Instrument	Criterion	Evidence
VAL-09	REQ19	Simulate the three predefined abnormal conditions (low level, overtemperature, and pump failure) and verify activation of the visual and audible alarms.	Direct inspection	Warning correctly activated for all three simulated conditions.	Photographic record or video.
VAL-10	REQ15, REQ16	During a test, verify the updating of temperature, level, and cycle-status variables on both the HMI and the IoT supervisory platform.	Network-traffic capture, platform logs	$\geq 99\%$ of the expected records received, with an update period ≤ 30 s.	Network traces and platform captures.
VAL-11	REQ11	Compare the reading of level sensor LIT-101 installed in TK-102 with a manual reference measurement at a minimum of three different levels.	LIT-101, ruler or graduated reference mark	Error $\leq 5\%$ of the measured range in all three measurements.	Reading table and photographic record.
VAL-12	REQ01 (PMV, Sec. 4.5)	Activate gearmotor M-101 and agitator AG-101 during mixture preparation in TK-102 and verify fluid homogeneity after a defined agitation period, repeating the test three times.	Direct visual inspection, stopwatch	Visually homogeneous mixture in all three tests within the defined agitation time.	Photographic record or video.

For thermal validation, setpoints of 35°C , 40°C , and 50°C were defined, together with a controlled disturbance generated by adding cold fluid. The defined metrics include steady-state error, overshoot, settling time, dosing error, recovery percentage, level-measurement error, and variables associated with communication and safety-system performance. The metrics planned for the validation process are presented in Table IV.

TABLE IV
METRICS PLANNED FOR SYSTEM VALIDATION.

Area	Metric	Unit	Calculation Method
Control	Steady-state error, overshoot, and settling time	%, $^{\circ}\text{C}$	$e_{ss} = \frac{[SP - T_{steady}]}{SP} \times 100$; $M_p = \frac{T_{max} - SP}{SP} \times 100$. Calculated from the $T(t)$ curve exported from the PLC/HMI.
Communications	Latency, message loss, and reconnection time	ms, %, s	Network-traffic capture using timestamps; comparison between the number of expected and received packets.
Dosing	Absolute and relative error of the dosed volume	mL, %	Comparison between the programmed volume based on activation time and the volume measured using a graduated cylinder over three repetitions.
Recovery	Water-recovery percentage	%	Ratio between the volume recovered in TK-103 and the total volume supplied to the cycle.
Instrumentation	Level-measurement error (LIT-101)	%	Comparison between the sensor reading and a manual reference measurement in TK-102 at a minimum of three levels.
Agitation	Mixture-homogenization time (M-101/AG-101)	min	Time measured from the beginning of agitation until a visually homogeneous mixture is achieved, over three repetitions.
Safety	Response time of the emergency stop and alarms	s	Timing or PLC event capture from activation until actuator shutdown or alarm generation.

The validation plan therefore establishes the procedures and acceptance criteria required for the subsequent evaluation of prototype performance against the requirements formulated during this stage of the project.

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